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Experiment-03

Study of I-V Characteristics, Verification of CVD Model and Implementation of Logic Gates of Diode

CSE251 - Electronic Devices and Circuits Lab

Objective

1. To study the current-voltage characteristics, i.e. I-V characteristics of silicon p-n junction dio

ify the Constant Voltage Drop (CVD) model of a diode

(e) Observe the I-V graph and capture it with a camera.

Task-02: Verification of CVD model

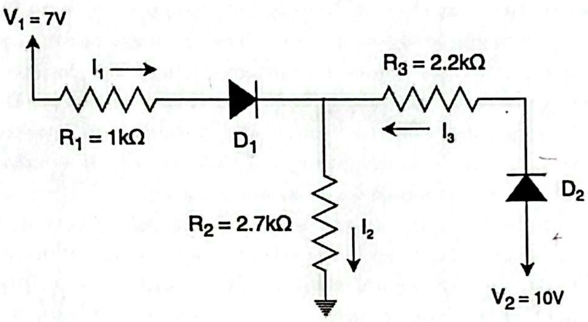


Figure 8: ***Circuit-2***

Procedure

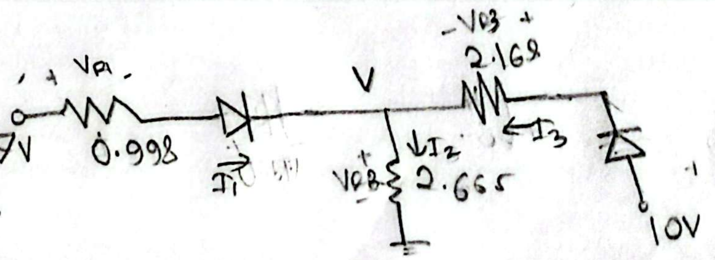
1. Use the Digital Multimeter to measure the resistances R_1, R_2, and R_3, as well as the forward voltage drops V_{DO_1} and V_{DO_2} of diodes D_1 and D_2, respectively. Fill in the tables below with the measured values.
2. Construct the ***Circuit-2*** given above.
3. Use the DC Power Supply to set $V_1 = 7V$ and $V_2 = 10V$ with 0.5A current limit
4. Now, measure the voltages across the resistances and fill out the tables.
5. For theoretical analysis, assume the diode follows the Constant Voltage Drop (CVD) model with a forward voltage drop of V_{DO}. Use the experimental values for both V_{DO} and the resistor in calculations to minimize error.

	V_{Do_1} (V)	V_{Do_2} (V)	R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	V_{R1} (V)	V_{R2} (V)	V_{R3} (V)	$I_1 = \frac{V_{R1}}{R_1}$ (mA)	$I_2 = \frac{V_{R2}}{R_2}$ (mA)	$I_3 = \frac{V_{R3}}{R_3}$ (mA)	D_1 (on/off)	D_2 (on/off)
Experimental	0.517	0.587	0.998	2.665	2.168	0.575	5.96	3.61	0.576	2.236	1.665	on	on
Theoretical			1	27	2.2	0.593	5.89	3.523	0.594	2.210	1.625	on	on

Percentage of error = $\left\lvert \frac{\text{Experimental} - \text{Theoretical}}{\text{Theoretical}} \right\rvert \times 100\%$	For I_1	For I_2	For I_3
	-3.03	1.17	2.46

$I_2 = \frac{V}{2.665}$

$I_3 = \frac{10 - V - 0.587}{2.168}$



so, we got

$\frac{7 - V - 0.517}{0.998} + \frac{10 - V - 0.587}{2.168} = \frac{V}{2.665}$

$V = 5.89V$

$I_1 + I_3 = I_2 \Rightarrow I_1 = \frac{7 - V - 0.517}{0.998}$

$V_{R1} = 7 - 5.89 - 0.517 = 0.593$ | $V_{R2} = 5.89$ | $V_{R3} = 3.523$

For OR Gate

Input Voltage, V_X (V)	Input Voltage, V_Y (V)	State of LED (On/Off)	Boolean Output (0 or 1)	Output Voltage, V_O (V) (LED Disconnected)
0V	0V	off	0	0V
0V	5V	on	1	4.12V
5V	0V	on	1	4.13
5V	5V	on	1	4.35V

For AND Gate

Input Voltage, V_X (V)	Input Voltage, V_Y (V)	State of LED (On/Off)	Boolean Output (0 or 1)	Output Voltage, V_O (V) (LED Disconnected)
0V	0V	off	0	0
0V	5V	off	0	0
5V	0V	off	0	0
5V	5V	on	1	5.05V

29-10-25

Signature of the lab faculty

Test Your Understanding

Answer the following questions:

1. $R = 1k\Omega$ was used in the experiment. If we use $R = 2.2k\Omega$, will there be any problem in observing the I-V characteristics plot? Explain briefly.
   Answer: We used the $1k\Omega$ Resistor to get the current, as $V=IR$, $I=\frac{V}{R}$, as $R=1$, $I=V$, now if we increase R, the graph will become compressed in the y axis, so the graph we observe would retain a similar shape but it wont be accurate, or it would be harder to interpret as y axis would become $I/2.2$ instead of I.

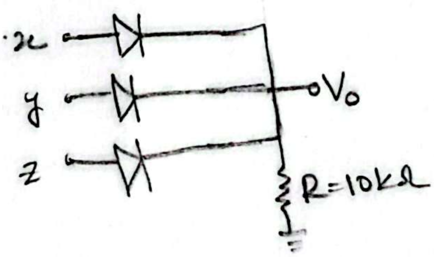
2. From the I-V characteristics of a diode that you obtained, which devices can be used to model the diode?
   Answer:

   In forward bias, we can model the diode with a voltage source of 0.7V and a resistor in series.

   In reverse bias, we can model the diode as an open circuit (as no current flows)

3. Compare the ideal diode model, CVD model and CVD + r model. Which model is better and why?
Answer: Ideal is the simplest model, there is no threshold voltage, no voltage drop and in forward bias there is perfect conduction (short circuit), in reverse bias, there's perfect insulation (open circuit). CVD model includes a constant threshold voltage which is if $V_D > V_{Threshold}$, its forward bias, else its reverse bias, rest of the characteristics is same as Ideal. CVD + r model has that constant threshold but there's also a resistance (gotten using an extra resistor), where in forward bias, voltage increase gradually rather than direct short circuit.
CVD + r is the best here as its most realistic and resembles a diode characteristics the most.

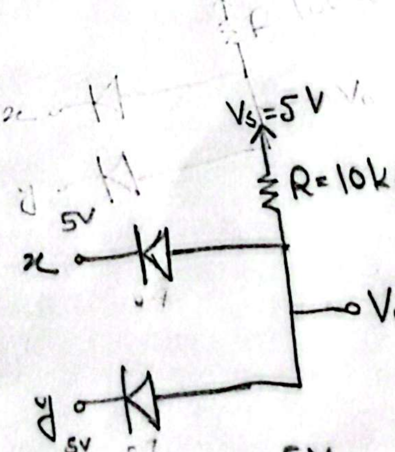
4. Design a 3-input OR gate using diodes and explain why a pull-down resistor is necessary. What happens if the resistor value is too large or too small?
Answer:



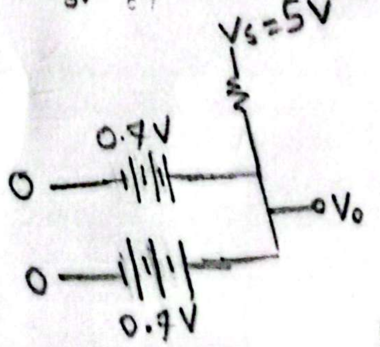
the pull down resistor ensures that when a voltage from x, y or z is high, V_o would be high too, if the resistor wasn't there, V_o would always be 0. as theres grounding.

If the resistor value is too high, it will take up too much power, leaving lower V_o value.
if the resistor value is too low, current flow through the diode is too high, so diode may get damaged.

5. Implement a 2-input AND gate using diodes. If the input high level is 5V and low level is 0V, calculate the actual output levels considering 0.7V diode drops.
Answer:



When both x and y = 5V since diode in reverse bias (x and y in negative side), diodes acts as open circuit and theres no current flow either so we get $V_o = 5V$ from V_s.

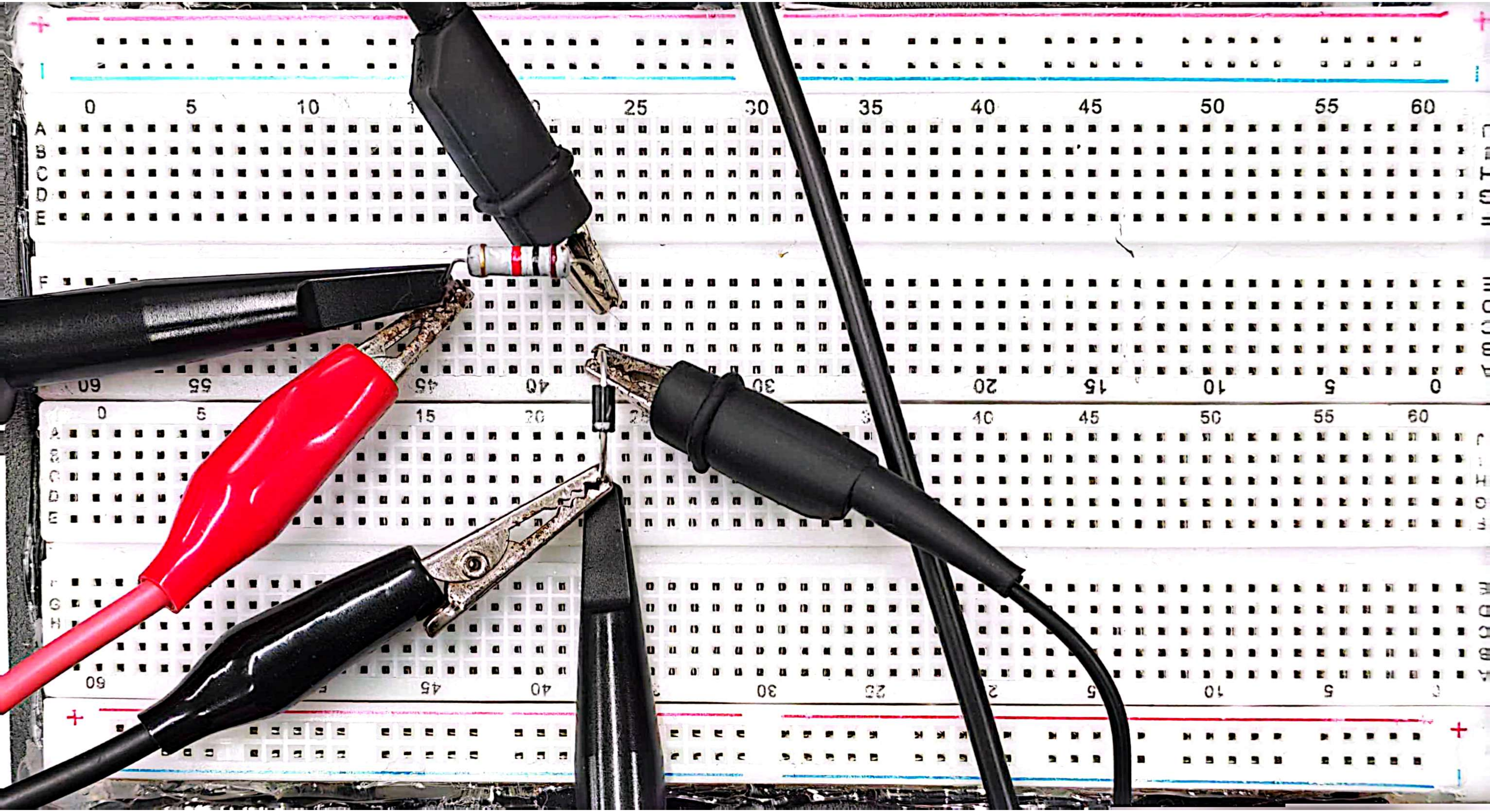


When either x or y or both is 0V, the diodes no longer acts as open circuit, they act as 0.7V source.
so $V_{out} = 0.7V$.

Reflection:

I learned about diode's IV characteristics, how it can be used to make logic gates and about it's different kind of models that can be used in circuits to represent a diode.
We had some confusion regarding when a diode is ON or OFF in Task 2, we simply asked sir to explain about it to us.

This experiment will help us to implement AND, OR gates using diodes in the future.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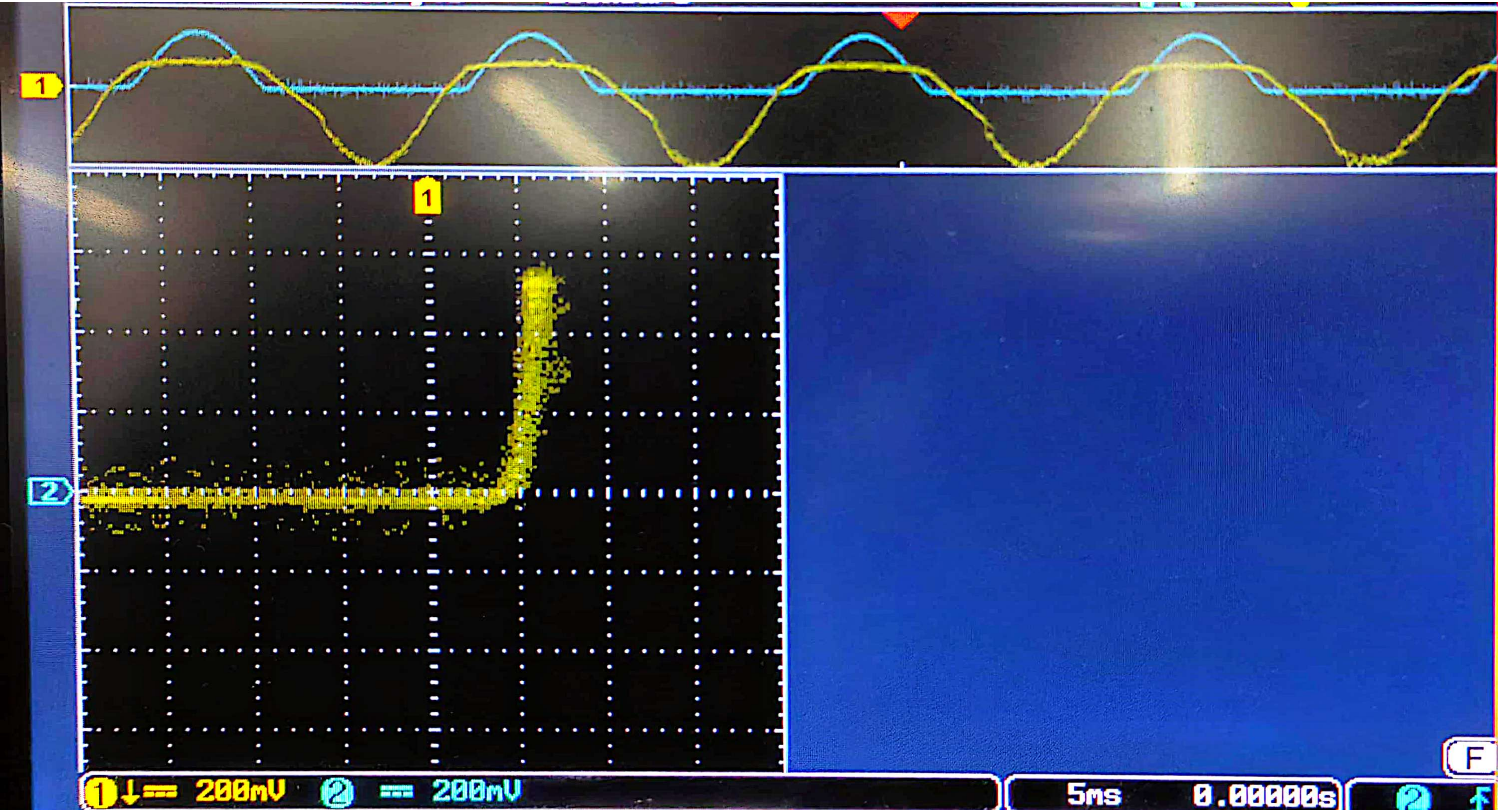
A B C D E
F
0 5 15 20 40 45 50 55 60
A B C D E
G H


1
1
2
200mV
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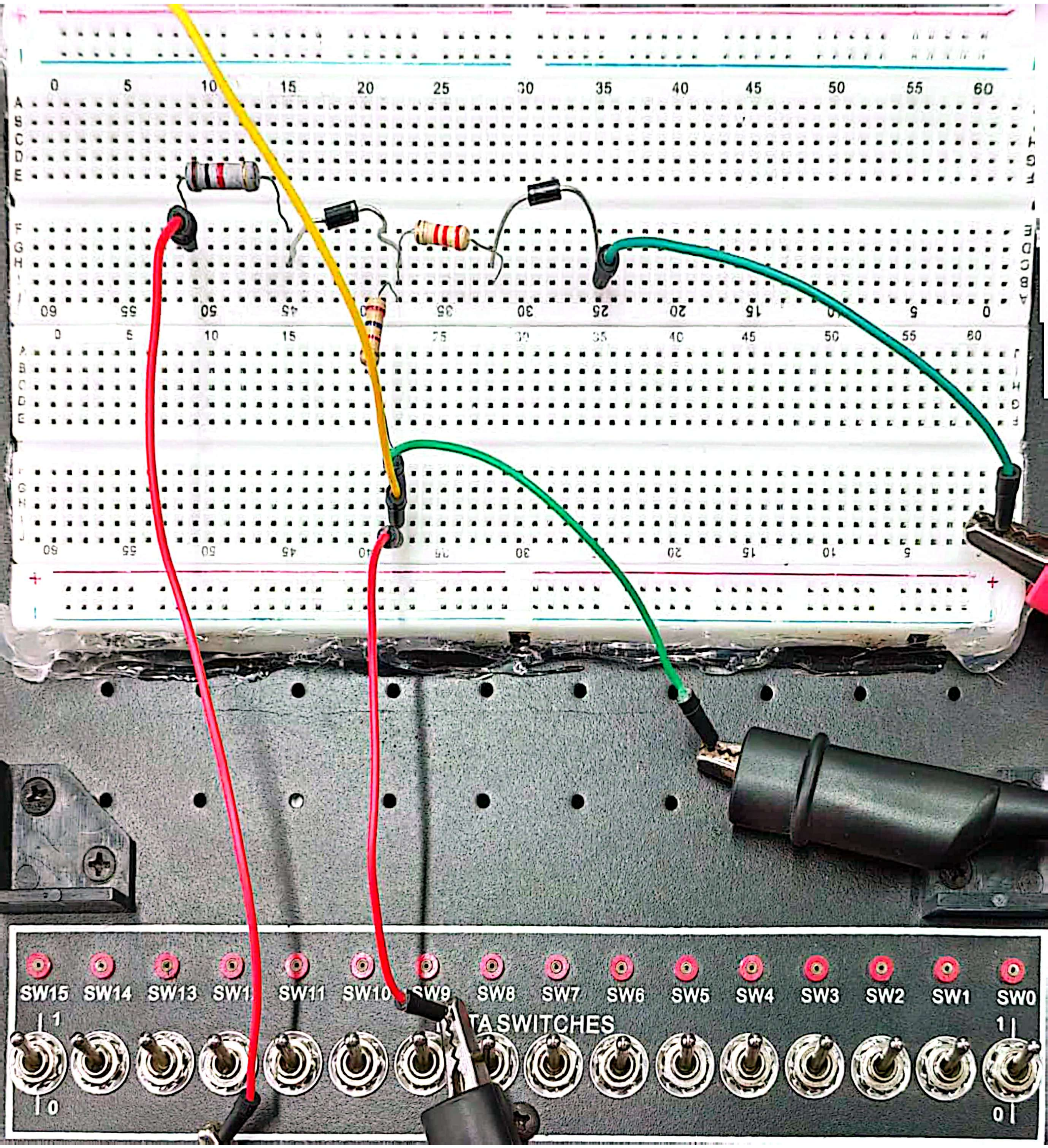
SW15 SW14 SW13 SW12 SW11 SW10 SW9 SW8 SW7 SW6 SW5 SW4 SW3 SW2 SW1 SW0
DATA SWITCHES